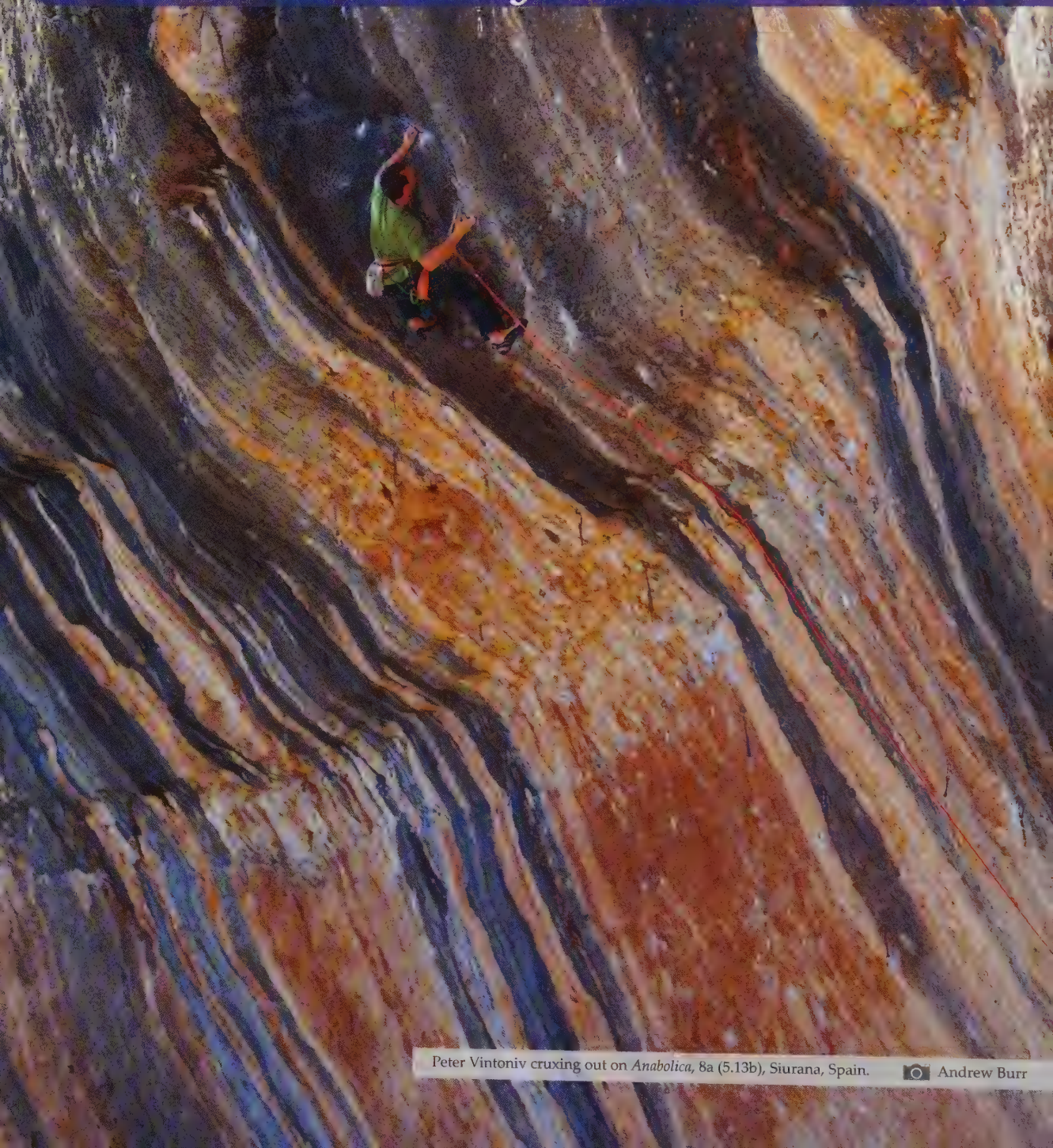


BUILDING A SEASONAL TRAINING PLAN AND OTHER TRAINING CONSIDERATIONS



Peter Vintoniv cruxing out on *Anabolica*, 8a (5.13b), Siurana, Spain.

 Andrew Burr

CHAPTER

10

"If we want to decrease the quantity of frustration and failure that we experience by not achieving our goals, we can do so only by trading out that type of pain for the other sort of pain — the sort of pain that comes from hard work and sacrifice."
— Bill Ramsey⁴

The previous chapters explained in detail how to execute each phase of the Rock Prodigy training program, presenting a selection of effective training routines that each climber can use to propel his or her climbing performance to new heights. The last step is to synthesize that information into a practical, easy to follow, day-to-day training plan. This training plan will describe specifically which training routines to perform on each day of the season, from day one of the Base-Fitness Phase, through the last day of the Rest Phase. A well-conceived schedule will guide the climber through each phase of the training cycle, maximizing improvement opportunities, while ensuring sufficient recovery to optimize progression and reduce the risk of injury. Finally, a detailed Seasonal Training Plan can provide tremendous encouragement when motivation is lacking. Think of this plan as a coach, unphased by the discomfort of daily training efforts, and stubbornly demanding maximum commitment from the athlete.

Ideally, an individualized Seasonal Training Plan should be sculpted to fit the specific needs of each climber. This chapter presents a variety of sample Seasonal Training Plans, along with tips for personalizing them. Select the sample plan that best applies to you, examine it closely, and then make initial adjustments to tailor it to your goals and desires. After using this prototype plan for a season, make further adjustments as experience dictates and as your strengths, weakness, and goals change. Continue iteratively adjusting this plan from season to season, experimenting with different training volume, recovery periods,

phase lengths, and phase timing to develop the optimal personal-training program.

Finally, this chapter briefly discusses a variety of alternative training approaches, including the pros and cons of each method. This information can be used to evaluate their suitability, and identify situations in which these approaches may be warranted.

BUILDING A SEASONAL TRAINING PLAN

Chapter 2: Goal Setting and Planning discussed the planning process in broad terms. Now with a comprehensive understanding of the purpose, theory, and application of each training phase, it's time to synthesize this information into a practical, individualized macro-cycle. Seasonal Training Plans should vary from climber to climber, based on physical traits, strengths and weaknesses, and training experience. Furthermore, even for a single climber, the macro-cycle should vary from season to season as the climber progresses, and in accordance with the climber's season-specific goals. A macro-cycle geared toward a bouldering project will look much different than one aimed at free-climbing El Cap.

The following discussion will present two example macro-cycles, tailored to climbers of different experience levels, then explore various tradeoffs to consider when adapting such plans to one's own abilities, experience, and goals. The plans presented below are far more detailed than the conceptual plan presented in the Figure on p39. Such detail is not always necessary, but it can be extremely helpful, espe-

"The brain is the most important muscle for climbing." — Wolfgang Gullich²

BUILDING A SEASONAL TRAINING PLAN

	Week	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Base Fitness Phase Strength Phase Power Phase Power Endurance Phase Performance Phase Rest Phase	-	Last Week of Performance Phase of Previous Season						
	-	Rest Phase of Previous Season						
	-	Day 1 *Outdoor Mileage (OM), 6 mod pitches *Skill acquisition						
	1	Su 1 *OM, 4 mod pitches *Skill acquisition	Mo 2 *Optional ARC, 2 x 20 min *OAE	Tu 3 *ARC 1: 2 x 20 min *Skill acquisition	We 4 *OAE	Th 5 *ARC 2: 2 x 25 min *Skill acquisition	Fr 6 *OAE	Sa 7 *OM, 7 mod pitches *Skill acquisition
	2	Su 8 *OM, 5 mod pitches *Skill acquisition	Mo 9 *OAE	Tu 10 *ARC 3: 2 x 30 min *Skill practice	We 11 *OAE	Th 12 *ARC 4: 2 x 30 min *Skill practice	Fr 13 *OAE	Sa 14 *OM, 8 mod pitches *Skill practice
	3	Su 15 *OM, 6 mod pitches *Skill practice	Mo 16 *OAE	Tu 17 *ARC 5: 3 x 25 min *Skill practice	We 18 *ARC 6: 3 x 25 min *Skill practice	Th 19 *ARC 7: 2 x 25 min *Stress-proofing	Fr 20 *OAE	Sa 21 *OM, 10 mod pitches *Stress-proofing
	4	Su 22 *OM, 8 mod pitches *Stress-proofing	Mo 23 *OAE	Tu 24 *ARC 8: 3 x 30 min *Stress-proofing	We 25 *ARC 9: 3 x 30 min *Stress-proofing	Th 26 *ARC 10: 2 x 30 min *Stress-proofing	Fr 27 *OAE	Sa 28 *Hangboard (HB) 1 *Supplemental Ex
	5	Su 29 *Optional OM, 4p *OAE *Skin care	Mo 30 *OAE	Tu 31 *HB 2 *SE	We 1 *Opt ARC 2x20 min *OAE *Skin care	Th 2 *OAE	Fr 3 *HB3 *SE	Sa 4 *Optional OM 4p *OAE *Skin care
	6	Su 5 *OAE	Mo 6 *HB 4 *SE	Tu 7 *Opt ARC: 2x20 min *OAE *Skin care	We 8 *OAE	Th 9 *HB5 *SE	Fr 10 *Opt ARC 2x20 min *OAE *Skin care	Sa 11 *OAE
	7	Su 12 *HB6 *SE	Mo 13 *Opt ARC 1 x 30min (Begin Tapering) *Skin care	Tu 14	We 15 *HB7 *SE	Th 16 *Skin care	Fr 17	Sa 18 *Warmup Boulder Ladder (WBL) 60m *Campus 15 min
	8	Su 19 *Skin Care	Mo 20	Tu 21 *Limit Boulder (LB) 90 min *SE	We 22 *Skin Care	Th 23 *WBL 60 min *Campus 20 min *SE	Fr 24 *Skin Care	Sa 25 *Skin Care
	9	Su 26 *LB Outdoors	Mo 27 *Skin Care	Tu 28 *WBL 60 min *Campus 25 min *SE	We 29 *Skin Care	Th 30	Fr 31 *LB 90 min *SE	Sa 1 *Skin Care
	10	*WBL 60 min *Linked Bouldering Circ (LBC), DC=2:4	*Skin Care		*WBL 50 min *LBC, Duty Cycle (DC) = 2:3; SE	*Skin Care		*Redpoint (RP) attempts on bouldery routes
	11	*Redpoint (RP) attempts on bouldery routes	*Skin Care		*WBL 50 min *LBC, DC = 2:2 *SE	*Skin Care		*RP Attempts *OM, 2 mod pitches
	12	*RP Attempts *OM, 2 mod pitches	*Skin Care		*WBL 40 min *Intervals, DC = 4:4 *SE	*Skin Care		*RP Attempts on pump routes *OM, 2 mod pitches
	13	*RP Attempts (pumpy routes) *OM, 2 mod pitches	*Skin Care		*WBL 40 min *Intervals, DC = 4:3 *SE	*Skin Care		*RP Attempts (pumpy routes) *OM, 2 mod pitches
	14	*RP/On Sight (OS) Attempts *OM, 2 mod pitches	*Skin Care	*RP Attempts (pumpy routes) *OM, 2 mod pitches	*RP/On Sight (OS) Attempts *OM, 2 mod pitches	*Skin Care	*RP/On Sight (OS) Attempts *OM, 2 mod pitches	*Skin Care
	15	*RP/OS Attempts *OM, 2 mod pitches	*Skin Care		*WBL 45 min	*Skin Care		*RP/OS Attempts *OM, 2 mod pitches
	16	*RP/OS Attempts *OM, 2 mod pitches		*OAE and/or Optional Cross-Training (OCT)		*OAE and/or OCT		*OAE and/or OCT
	17		*OAE and/or OCT		*OAE and/or OCT		*OAE and/or OCT	First Day of Next Season

Seasonal Training Plan for Novice Trainee focused on sport climbing. This example is tailored toward climbers with little training experience, climbing below the 5.12 level. It emphasizes the Skill Development-rich phases of Base-Fitness, Power-Endurance, and Performance, while minimizing the Strength and Power phases, since novice trainees experience significant improvement in these areas with minimal training volume. Optional ARC workouts are prescribed throughout the Strength Phase to further promote Skill Development. This schedule offers more opportunities for outdoor climbing than the advanced schedules. The activities for each day should be performed in the order they are listed. The day's primary activity is underlined. Note that this schedule includes a weeklong climbing vacation during Week 14.

BUILDING A SEASONAL TRAINING PLAN

Week	Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
-	<i>Last Week of Performance Phase of Previous Season</i>						
-	<i>Rest Phase of Previous Season</i>						
-							*Outdoor Mileage (OM) 8 mod pitches *Skill acquisition
1	*OM, 6 mod pitches *Skill acquisition	*OM, 6 mod pitches *Skill acquisition	*ARC 1: 2 x 30 min *Skill acquisition		*ARC 2: 2 x 35 min *Skill practice		*OM, 10 mod pitches *Skill practice
2	*OM, 8 mod pitches *Skill practice	*OAE	*ARC 3: 3 x 30 min *Stress-proofing	*ARC 4: 3 x 25 min *Stress-proofing	*ARC 5: 2 x 30 min *Stress-proofing		*Handboard (HB) 1 *Supplemental Ex
3	*Optional OM 4p *OAE *Skin care	*OAE	*HB 2 *SE	*OAE *Skin care	*OAE	*HB3 *SE	*Optional OM 4p *OAE *Skin care
4	*OAE	*HB 4 *SE	*OAE *Skin care	*OAE	*HB5 *SE	*OAE *Skin care	*OAE
5	*HB6 *SE	*Skin care	*OAE	*HB7 *SE	*Begin Tapering *Skin care		*HB8 *SE
6	*Skin care		*HB9 *SE	*Skin care		*Warmup Boulder Ladder (WBL) 60m *Campus 25m; SE	*Skin Care
7		*Limit Boulder (LB) 120 min *SE	*Skin Care	*WBL 60 min *Campus 30 min *SE	*Skin Care		*LB Outdoors
8	*Skin Care	*WBL 60 min *Campus 35 min *SE	*Skin Care		*WBL 60 min *Campus 40 min *SE	*Skin Care	*LB Outdoors (on boulders or routes)
9	*Skin Care	*WBL 60 min *Campus 45 min *SE	*Skin Care		*LB 120 min *SE	*Skin Care	*Redpoint (RP) attempts on bouldery routes
10	*Skin Care	*WBL 60 min *Linked Bouldering Circ. (LBC); DC=2.4	*Skin Care	*WBL 50 min *LBC, Duty Cycle (DC) = 2.3; SE	*Skin Care		*Redpoint (RP) attempts on bouldery routes
11	*Redpoint (RP) attempts on bouldery routes	*Skin Care		*WBL 50 min *LBC, DC = 2.2 *SE	*Skin Care		*RP Attempts *OM, 2 mod pitches
12	*RP Attempts *OM, 2 mod pitches	*Skin Care		*WBL 40 min *Intervals, DC = 4.4 *SE	*Skin Care		*RP Attempts on pumper routes *OM, 2 mod pitches
13	*RP Attempts (pumper routes) *OM, 2 mod pitches	*Skin Care		*WBL 40 min *Intervals, DC = 4.3 *SE	*Skin Care		*RP Attempts (pumper routes) *OM, 2 mod pitches
14	*RP Attempts (pumper routes) *OM, 2 mod pitches	*Skin Care		*WBL 40 min *Intervals, DC = 4.2 *SE	*Skin Care		*RP/On Sight (OS) Attempts *OM, 2 mod pitches
15	*RP/OS Attempts *OM, 2 mod pitches	*Skin Care		*WBL 45 min	*Skin Care		*RP/OS Attempts *OM, 2 mod pitches
16	*RP/OS Attempts *OM, 2 mod pitches						*OAE and/or OCT
17		*OAE and/or OCT		*OAE and/or OCT		*OAE and/or OCT	First Day of Next Season

Key:

HB - Hangboard Workout
Circuits - OAE - Optional Aerobic Exercise
OM - Outdoor Mileage
WBL - Warm-Up Boulder Ladder

DC - Duty Cycle (work-to-rest ratio)
M/mod - Moderate
OS - Onsite
ARC - Aerobic Restoration and Capillarity Training

m/min - Minutes
OCT - Optional Cross-Training
SE - Supplemental Exercises

LBC - Linked Bouldering
RP - Redpoint

Seasonal Training Plan for Experienced Trainee, emphasizing finger-strength training for sport climbing. This example is tailored to an athlete with several years of training experience, climbing at the 5.13 level. This schedule assumes the athlete has strong movement skills but performance-limiting finger strength. The schedule maximizes the Strength and Power phases (since advanced athletes require far more training volume to stimulate finger-strength gains) while minimizing Base-Fitness. This schedule requires greater commitment to indoor training. The activities for each day should be performed in the order they are listed. The day's primary activity is underlined.

BUILDING A SEASONAL TRAINING PLAN

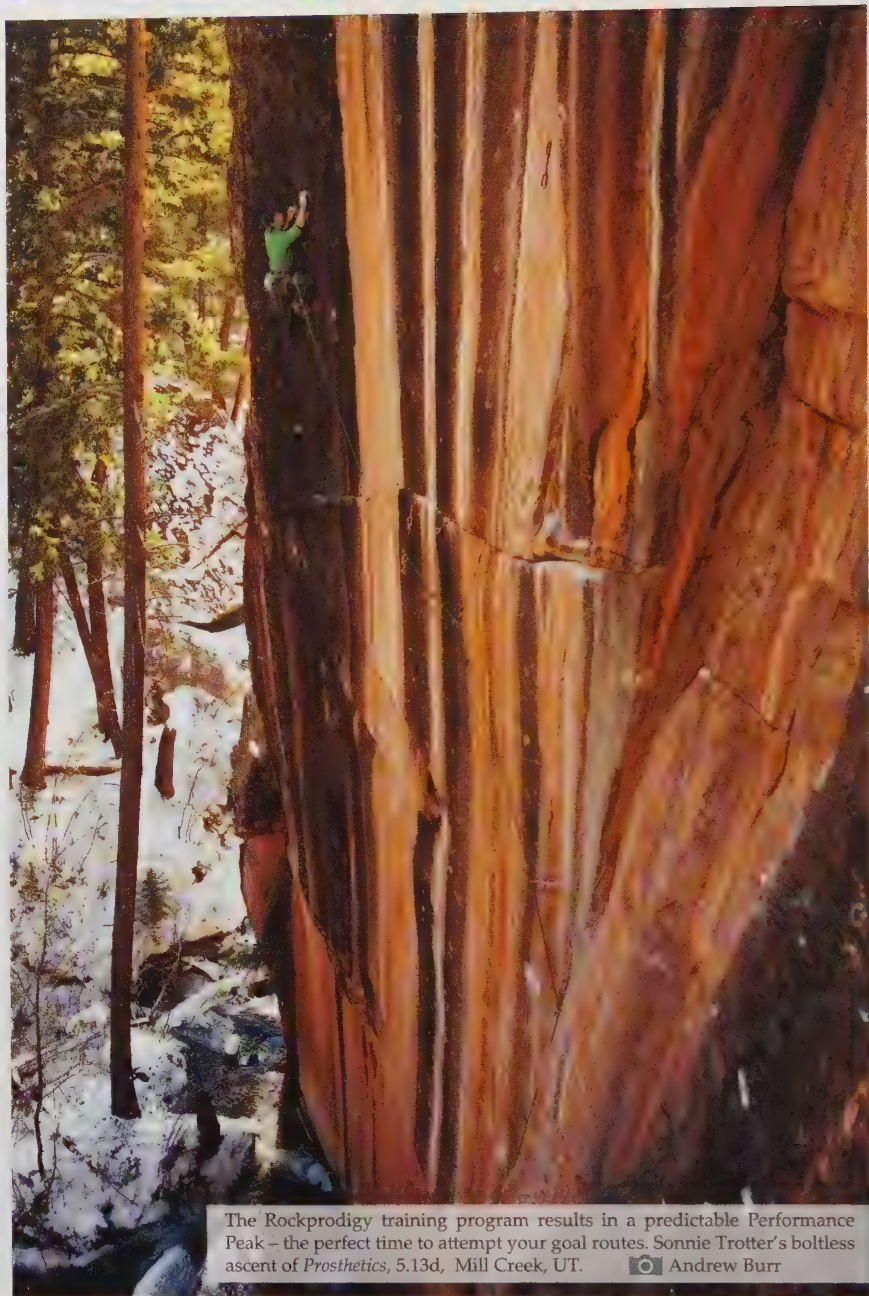
cially for those new to periodic training. These plans can be followed exactly, and neophytes should do just that (after selecting the most suitable plan). However, eventually it will be in each climber's best interest to diverge from these examples as experience and knowledge evolve. **Chapter 14: Traditional and Big Wall Free Climbing** and **Chapter 15: Bouldering** each present additional example plans specific to those styles of climbing.

TAILORING THE SEASONAL TRAINING PLAN

Create a draft schedule *before* beginning the macro-cycle. A firm schedule serves as a virtual coach, committing the climber to the program, and minimizing indecision that often leads down the path of least resistance. Once the schedule is drafted, changes should be well-considered and implemented at a time of rest, when the desire to avoid near-term discomfort is less likely to influence the decision. The following tips will help climbers tailor the above baseline Seasonal Training Plans to fit their individual physical attributes, address strengths and weaknesses, and train for specific goals:

INDIVIDUALIZING YOUR PLAN:

- **Less-Experienced Climbers:** Use relatively longer Base-Fitness, Power-Endurance, and Performance phases to maximize opportunities for Skill Development. Furthermore, these climbers should maximize opportunities to climb outdoors (where Skill Development is most specific) throughout the macro-cycle.
- **More-Experienced Climbers:** Use relatively longer Strength and Power phases since a greater training volume is required to stimulate adaptation in



The Rockprodigy training program results in a predictable Performance Peak – the perfect time to attempt your goal routes. Sonnie Trotter's boltless ascent of *Prosthetics*, 5.13d, Mill Creek, UT.  Andrew Burr

well-trained athletes. Furthermore, such climbers should minimize supplemental ARC workouts during the Strength Phase to facilitate maximum intensity during strength workouts.

- **Skill-Dependent Goals** (such as a difficult slab climb or technical testpiece): Use relatively longer Base-Fitness and Performance phases to maximize opportunities for skill acquisition, practice, and stress-proofing. These climbers should also increase outdoor skill practice sessions, especially later in the macro-cycle.

"I record everything in a training diary — you get stronger very gradually so it's useful to look back and see what you were doing in terms of quality and quantity in a set amount of time."

— Ben Moon¹

- **Finger Strength–Dependent Goals** (such as fingery bouldering or short, bouldery routes): Maximize the Strength and Power phases while drastically reducing the Base-Fitness and Power-Endurance phases as they are detrimental to pure power.

TIPS FOR MACRO-PLANNING:

- **Phase Lengths:** Ideally, phase lengths should be adjusted so that a phase ends just as the climber begins to plateau. If a plateau arrives earlier than expected, consider transitioning to the next phase ahead of schedule. Likewise, if there is no sign of a plateau at the end of a phase, it may be prudent to extend the phase, if doing so aligns with the climber's goals.
- **PE and Performance Phases:** Note that these phases are similar in nature. In fact, they could be considered two halves of the same phase. Focus more on indoor training early in the PE Phase, and more on outdoor performance climbing later in the phase.
- **Performance Peak Timing:** The Rock Prodigy program typically results in a *power peak* at the end of the Power Phase, followed by an *endurance peak* late in the Performance Phase. Plan performance goals accordingly. For example, plan to attempt boulder problems or bouldery routes during the power peak, and long, enduro redpoints or onsights during the endurance peak. As noted previously, attempting tweeky or powerful routes late in the season often leads to injury, and thus should be avoided.
- **Climbing Trips:** Road trips or other opportunities for plentiful outdoor climbing should be planned for the first two weeks of the Performance Phase. Those with the flexibility to choose when to use vacation time should arrange climbing trips for periods of optimal weather at choice crags, then plan the macro-cycle around that trip. Those with fixed vacation periods should schedule macro-cycles such that the Performance Phase coincides with maximal free time (doing so will create a ripple effect that impacts other seasons, so take that into account when deciding

when to start a macro-cycle).

- **Aerobic Exercise:** As discussed in **Chapter 8: Power-Endurance**, general aerobic endurance training has little direct benefit to rock-climbing performance. Such activities should be limited to the early phases of the macro-cycle, and even then only performed at low intensity to avoid draining effort away from climbing-specific training. Swimming in particular can further stress elbow and finger joints, and is bad for the skin, so should be avoided.

TIPS FOR MICRO-PLANNING:

- **Weekends:** Example plans are designed assuming Saturday and Sunday are days off from work. Climbers with other off days should adjust according to these guidelines:
 - If possible, avoid back-to-back outdoor *performance* days (back-to-back outdoor days during Base Fitness are acceptable).
 - Take two consecutive rest days after two consecutive climbing days.
 - Take one rest day after single climbing days.
 - Take two consecutive rest days (~70 hours) between high-intensity workouts such as hangboard, campus, or interval workouts. At the very least, attempt to schedule two consecutive rest days at least once every two weeks during Strength, Power, PE, and Performance phases. A relatively low-volume ARC workout can be performed on the first "rest" day.
- **Order of Exercises:** When performing different types of training on the same day, perform the more intense activity first (after a thorough warm-up). For example, during the Power-Endurance Phase, perform the various exercises in this order: 1) Warm-Up Boulder Ladder (including Hard Bouldering), 2) Linked Bouldering Circuit, 3) ARC set.
- **Order of Performances:** When climbing outdoors on consecutive days (such as over a two-day weekend), plan to perform more intense or powerful activities on the first day, and more endurance-oriented activities on the second day. For example, boulder or attempt difficult redpoints on day one; then attempt onsights or climb a multipitch trad route on day two. Alternatively, if one objective is more valued than another, attempt that objective first.

OUTDOOR TRAINING

In general, the training activities that make up the Seasonal Training Plan should be performed on the terrain specified in prior chapters (typically on an artificial climbing apparatus, or optionally outdoors, when noted). However, in some cases it is appropriate to perform outdoor training instead of indoor training, though care should be taken to tailor such climbing to the goals of that particular training phase (and the specific workout being replaced). The merits of specialized training were described at length in **Chapter 1: Introduction**, so the primary disadvantage of choosing to train outdoors is the rejection of those merits. Outdoor workouts are less quantifiable, less controlled, less thorough, less intense, and generally less effective than the exercises they replace.

However, there are some advantages to outdoor training, and a few specific reasons for doing it. Primarily, training on rock is very climbing specific (but only specific to the chosen rock type and route characteristics). Additionally, it sometimes offers the opportunity to make progress on a goal route while simultaneously accomplishing training objectives. In some cases, an indoor gym is not available, so outdoor climbing is the only option for certain types of training.

Certain long-term or seasonal goals warrant additional on-rock time. A climber preparing for a goal on an unfamiliar rock type or climbing style may be better prepared by training on that rock type. For example, if the goal is to free-climb Yosemite's El Capitan, it is essential to be skilled at slab, face, and crack climbing on granite. It may be impossible to accrue the necessary practice time without replacing some indoor training days with outdoor crag days. Another good case for training outdoors is the climber who struggles to adapt to outdoor climbing (and has a large discrepancy between indoor and outdoor climbing performances). In this case, further indoor training is unlikely to produce results on the rock anyway, so every effort must be made to overcome this disconnect between rock and plastic before focusing further effort on physical improvement.

When used, outdoor training should not completely replace all of the training exercises of a given phase. Outdoor training should be used intermittently; alternate outdoor workouts with recommended indoor training activities. This approach avoids excessive layoffs between like workouts. For example, consider the two-week schedules shown in the Figure at the top of the next page. These schedules demonstrate two options for incorporating outdoor bouldering with hangboard training.

Some objectives demand specific skill development that can only be accomplished on real rock. Slabmaster Justen Sjong finessing his way up *Town Crier*, 5.12+, Index, WA.  Andrew Burr



BUILDING A SEASONAL TRAINING PLAN

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Su	Mo	Tu	We	Th	Fr	Sa
						*Limit Boulder Outdoors 1 *Supplemental Ex
Su *OAE *Skin care	Mo *Hangboard 1 *SE	Tu *Opt ARC: 2x20 min *Skin care	We *OAE	Th *Limit Boulder Outdoors 2 *SE	Fr *OAE *Skin care	Sa *Hangboard 2 *SE
4 Days						
Su *Optimal OM: 4b *OAE *Skin care	Mo *OAE	Tu *Limit Boulder Outdoors 3 *SE	We *OAE *Skin care	Th *Hangboard 3 *SE	Fr *Opt ARC: 2x20 min *Skin care	Sa *OAE
4 Days						

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Su	Mo	Tu	We	Th	Fr	Sa
						*Limit Boulder Outdoors 1 *Supplemental Ex
Su *OAE *Skin care	Mo *Hangboard 1 *SE	Tu *Opt ARC: 2x20 min *Skin care	We *OAE	Th *Limit Boulder Outdoors 2 *SE	Fr *OAE *Skin care	Sa *Limit Boulder Outdoors 3 *SE
6 Days						
Su *OAE *Skin care	Mo *Hangboard 2 *SE	Tu *Opt ARC: 2x20 min *Skin care	We *OAE	Th *Hangboard 3 *SE	Fr *Opt ARC: 2x20 min *Skin care	Sa *OAE
2 Days						

Sample schedules incorporating outdoor training during the Strength Phase. The first plan is preferred because it uses a consistent four-day gap between hangboard workouts, while the second plan uses a six-day gap at times. Such a long gap makes it difficult to correctly progress the load between hangboard workouts.

The first schedule uses consistent four-day gaps between indoor hangboard workouts. The second schedule is less consistent, varying the spacing between hangboard workouts anywhere from two to six days. The latter approach makes it difficult to predict the proper training volume for each workout, reducing the chances of achieving progressive overload and reducing the effectiveness of the training.

Outdoor training days should not be random, "climb whatever comes to mind" cragging. Apply the same deliberate approach that is used for training; establish a goal for the day that supports the long-term season goals, make a plan to achieve the goal,

and execute the plan. Outdoor training days should be tailored to fit the training goals of the phase in which they occur. Follow the guidelines in the figure below for outdoor-climbing activities that are appropriate for each phase.

The timing of sets and repetitions of outdoor training can mirror those used in the recommended exercises (mostly applicable to the ARC and PE workouts), but this may be impractical. Given the general inefficiency of outdoor climbing, expect to spend more time at the crag to obtain a training volume similar to that accomplished during indoor training.

Phase	Typical Training Exercise	Comparable Outdoor Training
Base-Fitness	ARC	Outdoor Mileage (climb many, moderate-intensity pitches focusing on skill practice, breathing & relaxation)
Strength	Hangboard Training	Bouldering or "roped bouldering" (working powerful route cruxes). Emphasize longer cruxes with fingery holds.
Power	Campus or Limit Bouldering	Bouldering or "roped bouldering". Emphasize shorter cruxes with dynamic movements.
Power-Endurance	Linked Bouldering Circuits	Redpoint/onsight attempts on pumpy routes (similar in length to goal routes) -or- Interval Laps
Peak	Redpoint & Onsight Attempts	Same

Training Mediums: Rock vs. Plastic

In theory, any of the exercises described in this program can be performed outside on rock instead of on artificial structures. However, experience shows that indoor training is much more effective at producing physical improvements in strength and power (whereas outdoor training is more effective for Skill Development, mental training, and often power-endurance training). The reason for this is that strength and power training rely heavily on the willingness of the athlete to push right up against his or her physical limitations — an activity best performed in a safe, controlled environment. Plastic (and steel) training tools allow athletes to carefully control individual variables, ensuring that all the relevant principles of effective training are addressed. Rock is too random, unpredictable, and ultimately unquantifiable to facilitate optimal training.

For best results, maximize indoor training opportunities to the extent possible when focusing on strength and power. When focusing on Skill Development or mental training, seek opportunities to climb

on real rock. As an example, PE training can be accomplished successfully on either medium, so the choice hinges on the quality of the gyms *and* crags available. Crags with steep, tall, jug-filled walls, like those found at the Red River Gorge, are ideal for PE training. Likewise, a tall gym with good route-setting can be superior to many crags. Gyms also provide more variety, and thus options for hold type, training duration, and intensity. Either way, consider the tradeoff between specificity and the principles that rely on accurate quantification of training stress when selecting your training medium.



The right outdoor crag, such as the Red River Gorge's Motherlode, can provide better Power-Endurance training than many gyms.  Dan Brayack

"In the early days of sport climbing, training was thought to be 'uncool' among the climbers in our circle. 'A good climber doesn't need any training,' was the expert opinion. And in truth, up until that time, all you needed was good technique and the right attitude. But times had changed. The difficult routes were increasingly overhanging, so climbers were forced to think much more about their sport and the training that was necessary for it." — Alex Huber³

OTHER TRAINING CONSIDERATIONS

OTHER COMMON TRAINING ACTIVITIES

The Rock Prodigy program is proven to raise the level of performance of climbers of varying experience levels, body types, and abilities. Therefore, we have high confidence that it will be effective for every climber, if it is followed sincerely. Nevertheless, there are numerous other training activities that have their own merits, and it's sensible to consider them. The following discussion is included to acknowledge these activi-



Roped bouldering on a fingery route such as this can be a reasonable substitute for indoor finger strength training. Steve Maisch working out on *Calamity Jane*, 5.13b, Wild Iris, WY.  Brendan Nicholson

ties and present the reasons for excluding them from the program. Those climbers who find these activities beneficial can incorporate them into their individualized Seasonal Training Plan.

SYSTEM TRAINING

System training is a form of structured bouldering that uses complex movements to apply a climbing-specific, controllable, repeatable, and somewhat quantifiable load to the entire body. System training was developed by the Germans Alex and Thomas Huber along with their coach Rudi Klausner³. The exercise requires a specially designed “system wall” that has identical, regularly spaced holds, so that exact movements can be repeated up and down the wall. The climber then performs “sets” by climbing up and down the wall, utilizing one specific grip at a time until the desired number of repetitions is performed. This training was intended as an improvement to campusing because it involves the whole body, “*through a perfect interplay of all the muscles in the chain between the fin-*

gertips and the toes.”³ System training can be used for strength or power training by adjusting the number of repetitions performed and intensity of each repetition (by adding weight to the climber or reducing hold size). This activity has several disadvantages including:

- Lack of muscle isolation
- Reduced ability to progress the load
- Reduced ability to quantify the load
- Reduced intensity

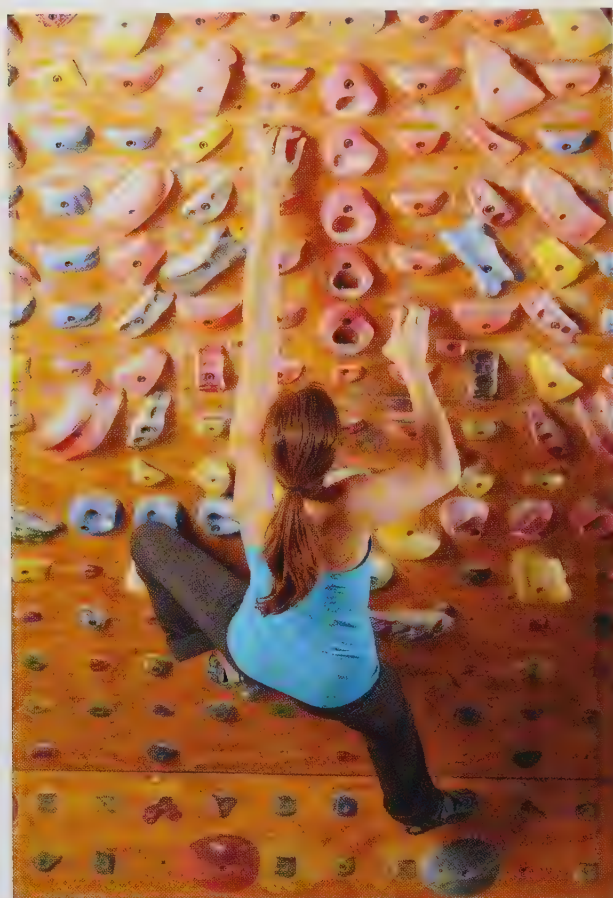
The primary disadvantage of this activity is the lack of muscle isolation. By including full-body movements, many potential failure points (finger flexors, biceps, latissimus dorsi, shoulders, core) are introduced. It is unlikely each of these links in the chain will achieve failure at the same point in the exercises. Physically, the fingers are the weak link for most climbers. Every effort must be made to improve this weakness, and that means the tools and methods best suited to improving finger strength should be used. Hangboarding isolates the finger flexors (and supporting structures) better than any other training tool, making it the best choice for those determined to improve finger strength. Campusing is more plyometric and dynamic, capitalizing on the stretch reflex to maximize muscle recruitment, while improving functional muscle speed, and thus power. Additionally, the campus board minimizes technical complexity, facilitating the enhanced arousal and commitment required of effective power training.

Furthermore, adjusting the resistance during system training is problematic. Weight vests can be used to add weight, but these cause unnatural movement, thus reducing specificity; meanwhile, predictably subtracting weight becomes virtually impossible, because the pulley system used to remove weight during hangboard training would be impractical on a tall, steeply overhanging system wall. Additionally, the load placed on the feet from rep to rep, set to set, and day to day can vary as technique varies, so it becomes difficult to control and quantify the load placed on the fingers (and the rest of the arm-shoulder-back-core chain). Therefore, it becomes impossible to carefully and regularly overload the finger flexors, and progressively increase that loading over time as the fingers adapt to training.

All of these detrimental factors combined make system training much less effective for improving finger *strength* than targeted hangboard training. On the other hand, for a climber who has adequate finger strength but lacks other strength for climbing move-

BUILDING A SEASONAL TRAINING PLAN

ment, system training may be appropriate (this would be a rare problem indeed). Most climbers have the upper-arm and back strength to perform individual pulls between good holds, but lack the endurance in these muscles to perform them repeatedly. One solution to this problem is to perform lock-off laps (described in [Chapter 6: Strength](#)) during the Power-Endurance Phase, in addition to performing highly effective hangboard training. Lock-off laps are a form of system training that de-emphasizes finger strength to instead isolate pull strength.



A System Wall can be used to train sport-specific strength with complex movements.

COMPLEX TRAINING

Complex training is an approach to strength training that was buzzing around climbing circles in the mid-2000s (though it has been around much longer in other sports). It involves combining strength exercises with plyometric exercises, and has been shown to improve speed and power output in large muscle groups (the legs, primarily). Typically, a set of strength-oriented lifts is performed followed by a short rest, then a set of a comparable plyometric exercise. For example, a basketball player (who needs to jump high) would

perform eight reps of squats, rest 90 seconds, then perform eight box jumps. This would be repeated for three total sets.

While known to be effective for many sports, there has never been a logical application of complex training to climbing. It has been argued that campus-ing is a form of complex training, and it is the most likely candidate for a means to adapt complex training to climbing. In this scenario, a climber might perform one weighted basic ladder up the board at low speed, followed by an unweighted max ladder at high speed (See [Chapter 7: Power](#) for descriptions of these exercises). This would provide the combination of a low-speed/high-intensity exercise followed by a high-speed/low-intensity exercise. The authors have experimented with combinations of hangboard and campus exercises as a form of complex training, but concluded that the results were no better than when these methods are used separately. Still, for those who would like to try it, the “*Combined Hangboard and Campus Board Maintenance Workout*” presented at the end of this chapter is a good baseline complex training routine.

MUSCLE CONFUSION

Numerous fad training programs are sweeping the globe (Crossfit, P90X, Insanity, etc.) that all essentially follow the same concept, that of *muscle confusion*. This concept is based on the principle of variability — that the athlete should vary exercises regularly and substantially to avoid training plateaus. Many folks swear by these programs for general fitness, but these are not specific to rock climbing and should not be expected to improve climbing performance. While they can help improve general fitness, these routines siphon training time and energy away from climbing, while adding muscle mass where it isn’t needed.

Still, many climbers argue that a similar approach could be adapted to climbing training, by using climbing-specific training apparatuses while varying the exercises much more frequently. This is not recommended. Climbing itself is already a form of muscle confusion. Routes and boulder problems bombard the climber with widely varying movements and intensities. Controlled, progressive, repetitive movement is extremely rare on the rock, or even on artificial walls. Climbing keeps the body in a constant state of confusion, so what it really needs is more regularity (another equally vital principle of effective physical training). The relatively few times within a climbing season that a climber performs targeted training is the only time

that the body experiences controllable, repeatable, quantifiable, and progressive loading. It is essential that these “islands of sanity” be maintained in what is otherwise a season of madness, in order to ensure steady, reliable adaptation and improvement.

AEROBIC EXERCISE

Many climbers include some form of aerobic exercise in their training. Aerobic exercise can be extremely enjoyable, and in moderation, good for overall wellness. However, the direct benefits to climbing performance are debatable. General aerobic fitness has little direct impact to performance on the rock. As discussed at length in **Chapter 5: Base-Fitness** and **Chapter 8: Power-Endurance**, climbing endurance is nearly always limited by the ability of the muscles in the forearms to produce ATP and remove waste products. Enhanced cardiovascular capacity has little bearing on these processes. That said, aerobic exercise can be helpful to climbers for these purposes:

- Establishing a basic level of fitness prior to beginning sport-specific training
- Maintaining healthy body composition
- Preparing climbers for extended multipitch climbs or strenuous approaches
- Preparing climbers for shorter routes at high altitude
- Improving general well-being and sanity

Some have theorized that aerobic exercise can help flush waste products out of the muscles between intense climbing or training days by increasing heart rate and blood flow. This may be true to an extent, but low-volume sport-specific exercise is far better suited to this purpose (such as short-duration ARC training).

ALTERNATIVE TRAINING APPROACHES: NON-LINEAR PERIODIZATION AND MAINTENANCE PROGRAMS

The training program described in this book utilizes a combination of *Linear Periodization* (LP) and *Non-Linear Periodization* (NLP), as explained in **Chapter 4: Foundations of Physical Training**. Whether it is deliberate or not, many climbers elect to use a Non-Linear Periodization approach exclusively, in which

all components of climbing fitness (strength, power, endurance) are trained simultaneously rather than separately in relatively distinct phases. With this approach, a weekly training schedule would include assorted workouts that target each component at least once during the weekly micro-cycle. Drawing from the exercises described here, consider the representative NLP schedule shown in the Figure at the bottom of the page.

Each of the fitness components is trained regularly in an effort to maintain all of them at a high level indefinitely.

This approach is appealing to some climbers because it facilitates uninterrupted outdoor climbing, avoiding the periods of exclusive indoor training that accompany the LP approach. The downside of NLP is that it results in a somewhat reduced level of peak fitness than LP. The LP approach delivers relatively predictable performance peaks that are ideal for those with limited on-the-rock climbing time, so that they can plan climbing vacations, etc. to coincide with peak fitness. On the other hand, climbers who have the desire and means to climb year-round, and who wish to do so at a consistent level, might prefer NLP.

NLP is only recommended as a piece of a larger LP approach. For those who wish to climb at a high level all the time, the unfortunate reality is that the human body responds cyclically to all training, both LP and NLP. In other words, regardless of the *training*, all climbing *performance* will be periodic, whether climbers like it or not. Therefore, the athlete should decide if he or she would like to control the timing of that performance. Those that choose NLP are choosing to take performance peaks and valleys as they come (seemingly at random).

This unpredictability of climbing fitness may not greatly impact newer climbers, who will experience steady progress as their skill improves, regardless of the approach. However, over time, this can be very frustrating, as the athlete is unable to count on steady improvement or predictable performance, with the good days and bad days seemingly uncorrelated to her training efforts.

A form of NLP that can be beneficial is the so-called *maintenance program*, which is used to main-

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
<i>Su</i> STRENGTH: *Hangboard	<i>Mo</i> BASE-FITNESS: *ARC	<i>Tu</i> *OAE	<i>We</i> POWER-ENDUR: *Linked Bouldering Circuits	<i>Th</i> *OAE	<i>Fr</i> POWER: *Campus Board	<i>Sa</i> *OAE

Representative Non-Linear Periodization training schedule.

STRENGTH AND POWER MAINTENANCE ROUTINE

WARM-UP:

- Low-intensity ARC traversing (20 to 30 minutes)
- Bodyweight (or less) dead hangs on the hangboard:
Three reps for three grip positions, 7/3 rep timing

EXERCISE 1: HANGBOARD ROUTINE:

- Perform seven-second reps with three seconds' rest as prescribed below for one grip position.
- The load performed during the first set of each grip for the last hangboard workout of the Strength Phase is the baseline load for this workout.

<u>Set</u>	<u>Reps</u>	<u>Load</u>
1	6	Baseline
	Rest 3:00	
2	6	Baseline + 10 lb.
	Rest 3:00	

EXERCISE 2: CAMPUS BOARD MAX LADDER:

<u>Set</u>	<u>Exercise</u>
1	Max Ladder, Leading Right
	Rest 1:30
2	Max Ladder, Leading Left
	Rest 3:00

REPEAT:

- Repeat **Exercise 1: Hangboard Dead Hangs** for the next grip position
- Repeat **Exercise 2: Campus Board Max Ladder**
- Continue until six grip positions have been completed.

SUMMARY:

The entire workout will consist of six circuits, with each circuit consisting of two sets on the hangboard and two sets on the Campus Board, performed in that order. Three minutes of rest should be taken between every set and every circuit.

COOLDOWN:

- Low-intensity ARC traversing (20 to 30 minutes).

tain and extend peak fitness after it has been achieved through an LP macro-cycle. During the Performance Phase, the climber makes every effort to perform on the rock as much as possible. However, performance climbing is poor training, and it is quite possible to rapidly lose fitness, especially with limited climbing days. Near the start of the Performance Phase, strength and power are initially high, but they begin to wane by the fourth week. Therefore, regular strength and power training during the Performance Phase are needed to sustain performance.

The best exercise for maintaining strength and power during the Performance Phase is Limit Bouldering on problems that contain both fingery and dynamic elements. This will provide a combination of climbing-specific strength and power stimuli. An effort should be made to select problems that are similar to the climber's goal routes to further increase specificity. The Rock Prodigy program includes these maintenance exercises in the form of the Warmup Boul-

der Ladder and Hard Bouldering included within the Linked Bouldering Circuit and Interval workouts of the PE and Performance phases. However, it may be desirable to further increase these exercises' emphasis in one of the following ways: 1) Increase the volume of Limit Bouldering within these workouts to train for more-powerful goal routes, 2) Increase the number of Limit Bouldering workouts to extend the duration of the Performance Phase. To accommodate these situations, a representative two-week maintenance schedule is shown in the Figure below for a climber who is able to perform outside on the weekends.

Limit Bouldering is the preferred maintenance training activity, but you can use other exercises as well. A combination of hangboard and Campus Board training is also effective (especially for those who don't have ready midweek access to bouldering). See the Figure on the opposite page for details on this workout.

Sunday	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Su PERFORM: *RP Attempts *OM, 2 mod pitches	Mo	Tu POWER: *Limit Boulder	We	Th POWER: *Limit Boulder	Fr	Sa PERFORM: *RP Attempts *OM, 2 mod pitches
Su PERFORM: *RP Attempts *OM, 2 mod pitches			We POWER: *Limit Boulder		Fr PERFORM: *RP Attempts *OM, 2 mod pitches	Sa

Representative maintenance training schedule for use during the Performance Phase.

SUMMARY

- Committing to a detailed Seasonal Training Plan is an essential part of the Rock Prodigy training program, and key to maximizing improvement.
- Ultimately, the Seasonal Training Plan should be tailored to each individual climber, and should vary from season to season as goals change. Initially, use one of the example plans provided, then make small adjustments as your knowledge and experience improve.
- Adjust the training emphasis of the plan in accordance with your training experience and the season's goals, by varying phase lengths and favoring indoor or outdoor training.
- Include performance events in the Seasonal Training Plan, accounting for periods of peak fitness, opportunities for maximal outdoor climbing, and optimal conditions.
- Typically, indoor training is superior for achieving physical improvements, while outdoor training is superior for achieving technical improvements. To be effective, outdoor training must be structured and tailored to the training objectives of the current phase.
- System training is another popular method for training strength and/or power.
- Aerobic exercise does little to benefit climbing performance in most cases, but it can be helpful in some situations.
- Complex training can be used as a form of Non-Linear Periodization or maintenance training, to extend a performance peak.